IEC 63035

Edition 1.0 2017-06

INTERNATIONAL STANDARD

MIDI (musical instrument digital interface) specification 1.0 (Abridged Edition, 2015)

IEC 63035:2017-06(en)

IEC Central Office
3, rue de Varembé
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

About the IEC
The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications
The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

IEC Catalogue - webstore.iec.ch/catalogue
The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

IEC publications search - www.iec.ch/searchpub
The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,…). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished
Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

Electropedia - www.electropedia.org
The world's leading online dictionary of electronic and electrical terms containing 20 000 terms and definitions in English and French, with equivalent terms in 16 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

IEC Glossary - std.iec.ch/glossary
65 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

IEC Customer Service Centre - webstore.iec.ch/csc
If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: csc@iec.ch.

IEC 63035

Edition 1.0 2017-06

INTERNATIONAL STANDARD

MIDI (musical instrument digital interface) specification 1.0 (Abridged Edition, 2015)

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 33.160.30; 35.040.01; 35.200

ISBN 978-2-8322-4355-8

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

INTERNATIONAL ELECTROTECHNICAL COMMISSION

____

MIDI (MUSICAL INSTRUMENT DIGITAL INTERFACE) SPECIFICATION 1.0 (Abridged Edition, 2015)

FOREWORD

1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
6) All users should ensure that they have the latest edition of this publication.
7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 63035 has been prepared by IEC technical committee 100: Audio, video and multimedia systems and equipment.

The text of this standard is based on the following documents:

CDV	Report on voting
100/2597/CDV	100/2858/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

IEC 63035 contains the same first 8 pages as in the MIDI 1.0 Detailed Specification (the original core specification text) published by the MIDI Manufacturers Association (MMA). These are included within this standard as Clauses 1 to 4. This specification was submitted to the IEC under the auspices of a special agreement between the IEC and the MMA.

The MMA is a non-profit corporation that serves as a support organization and forum for the advancement and adoption of MIDI technology (along with the Association of Musical Electronics Industry, or AMEI, in Japan).

The MIDI 1.0 technology dates back to 1983 when the protocol and electrical specification comprised 8 pages and the majority of the message identifiers were not yet defined. Over the subsequent years, the MMA and AMEI determined consensus of the worldwide MIDI industry, and defined numerous additional messages (via Confirmation of Approval documents), as well as many Recommended Practices for the use of MIDI technology, all the while maintaining MIDI as "1.0" (meaning that no significant changes were made to the initial specification).

The MMA documentation for MIDI 1.0 now encompasses more than 50 different documents in print or on the World Wide Web. This standard contains the same first 8 pages as in the MMA's MIDI 1.0 Detailed Specification but does not contain all of the subsequent information developed by MMA/AMEI. Rather, this document contains a complete listing (with basic description) of all defined MIDI messages to date, with references to the appropriate MMA documentation. Companies that want to implement MIDI technology are advised to also consult the MMA documentation that is listed in the Biography.

Although the MIDI 1.0 Detailed Specification includes an electrical connection specification ("MIDI-DIN"), other transports (USB, Firewire, etc.) have also been approved by MMA/AMEI for use with MIDI Protocol. For details and documentation of approved physical transports, please contact the MIDI Manufacturers Association.

The term "MIDI" is known all around the world as referring to the technology which is defined in the MMA/AMEI documents, and so should not be used for any other purpose. Companies that implement MIDI technology in their products in compliance with MMA specifications may use the term MIDI to describe their products, but may not use the term to describe any extensions or enhancements that are not defined by MMA/AMEI. Only MMA/AMEI can define the messages, transport payloads, and Recommend Practices which are promoted as "MIDI" so as to prevent any dilution and confusion of the meaning of "MIDI". Implementers of MIDI technology should consult MMA and/or AMEI (depending on the relevant market) for specific trademark usage policies.

MIDI (MUSICAL INSTRUMENT DIGITAL INTERFACE) SPECIFICATION 1.0 (Abridged Edition, 2015)

1 Scope

This International Standard specifies a hardware and software specification which makes it possible to exchange symbolic music and control information between different musical instruments or other devices such as sequencers, computers, lighting controllers, mixers, etc. using MIDI technology (musical instrument digital interface).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60130-9, *Connectors for frequencies below 3 MHz - Part 9: Circular connectors for radio and associated sound equipment*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1
velocity
parameter which typically changes the intensity and resultant volume of the note that is being played and varies according to the force applied

Note 1 to entry: Velocity is used as Key Velocity as in a piano key.

3.2
aftertouch
parameter that measures the level of intensity applied to a note after it has been played and continues to be depressed

Note 1 to entry: Typically, Aftertouch is useful for adding vibrato or tremolo effects to a sound in much the same way that a violin can add volume or pitch changes to a sustained note using finger vibrato or additional bowing intensity.

3.3
modulation wheel
wheel controller found on synthesizers that players can use to progressively introduce modulation depth to a sound

3.4
pitch wheel
wheel type device, normally found to the left of a synthesizer keyboard, used to manipulate the pitch of a played note or notes

3.5
pitch bend
activity or message, generally initiated by a pitch wheel, that smoothly raises and/or lowers the pitch of note or chord

3.6
oscillator
circuitry or software program that generates the kernel of a synthesizer sound

Note 1 to entry: In the early days, oscillators generated fairly basic sound types (sawtooth, square, pulse etc). In modern synthesizer engines, oscillators can be driven by myriad waveforms and samples.

3.7
pan
parameter that specifies the location of a sound within the stereo field

4 General

4.1 Hardware

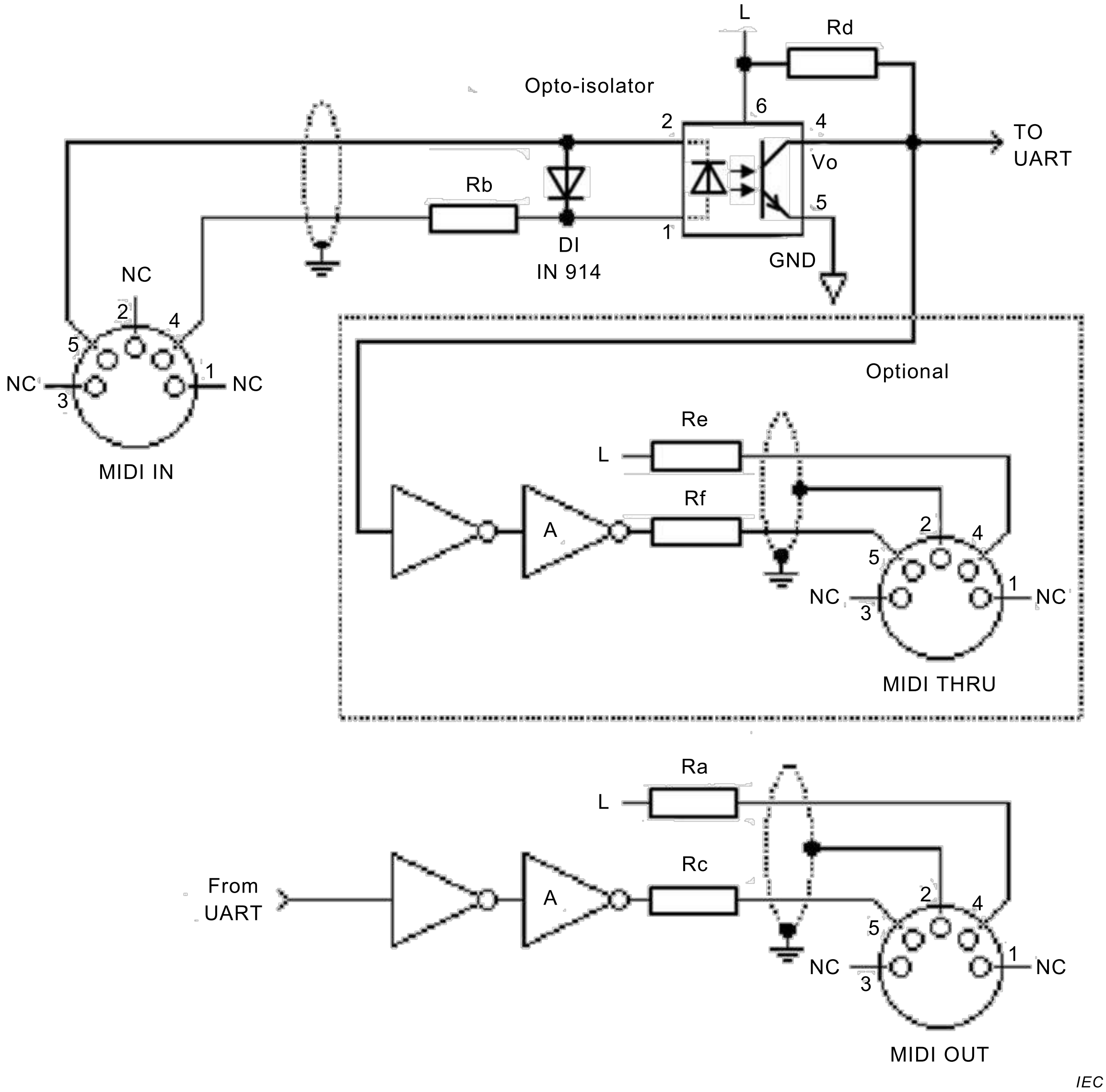
The hardware MIDI interface operates at 31,25 × (1 ± 1 %) kBd asynchronous, with a start bit, 8 data bits (D0 to D7), and a stop bit. This makes a total of 10 bits for a period of 320 µs per serial byte. The start bit is a logical 0 (current on) and the stop bit is a logical 1 (current off). Bytes are sent LSB first.

Circuit: (See Figure 1). 5 mA current loop type. Logical 0 is current ON. One output shall drive one and only one input. To avoid ground loops, and subsequent data errors, the transmitter circuitry and receiver circuitry are internally separated by an opto-isolator (a light emitting diode and a photo sensor which share a single, sealed package). The receiver shall require less than 5 mA to turn on. Rise and fall times should be less than 2 µs.

Connectors: DIN 5 pin (180°) female panel mount receptacle which is specified in IEC 60130-9 as type designation IEC-04. The connectors shall be labelled "MIDI IN" and "MIDI OUT". Note that pins 1 and 3 are not used, and should be left unconnected in the receiver and transmitter. Pin 2 of the MIDI In connector should also be left unconnected.

The grounding shield connector on the MIDI jacks should not be connected to any circuit or chassis ground.

When MIDI Thru information is obtained from a MIDI In signal, transmission may occasionally be performed incorrectly due to signal degradation (caused by the response time of the opto-isolator) between the rising and falling edges of the square wave. These timing errors will tend to add up in the "wrong direction" as more devices are chained between MIDI Thru and MIDI In jacks. The result is that, regardless of circuit quality, there is a limit to the number of devices which can be chained (series-connected) in this fashion.



Key

Components		**Supplies**
Ra, Rb, Rc, Re, Rf	resistor R = 220 Ω	L +5 V
Rd	resistor R = 280 Ω	

NOTE 1 Opto-isolator shown is Sharp PC-900. (HP 6N138 or other opto-isolator can be used with appropriate changes.)

NOTE 2 Gates "A" are IC or transistor.

NOTE 3 Resistors are 5 %.

Figure 1 – MIDI standard hardware

Cables shall have a maximum length of 15 m, and shall be terminated on each end by a corresponding 5-pin DIN male plug which is specified in IEC 60130-9 as type designation IEC-03. The cable shall be shielded twisted pair, with the shield connected to pin 2 at both ends.

A MIDI Thru output may be provided if needed, which provides a direct copy of data coming in MIDI In. For long chain lengths (more than three instruments), higher-speed opto-isolators should help to avoid additive rise/fall time errors which affect pulse width duty cycle.

4.2 Data format

MIDI communication is achieved through multi-byte "messages" consisting of one Status byte followed by one or two Data bytes. Real-Time and Exclusive messages are an exception.

A MIDI-equipped instrument typically contains a receiver and a transmitter. Some instruments may contain only a receiver or only a transmitter. A receiver accepts messages in MIDI format and executes MIDI commands. It consists of an opto-isolator, Universal Asynchronous Receiver/Transmitter (UART), and any other hardware needed to perform the intended functions. A transmitter originates messages in MIDI format, and transmits them by way of a UART and line driver.

MIDI makes it possible for a user of MIDI-compatible equipment to expand the number of instruments in a music system and to change system configurations to meet changing requirements.

MIDI messages are sent over any of 16 channels which are used for a variety of performance information. There are five major types of MIDI messages: Channel Voice, Channel Mode, System Common, System Real-Time and System Exclusive.

A MIDI event is transmitted as a "message" and consists of one or more bytes. Figure 2 to Figure 5 show the structure and classification of MIDI data.

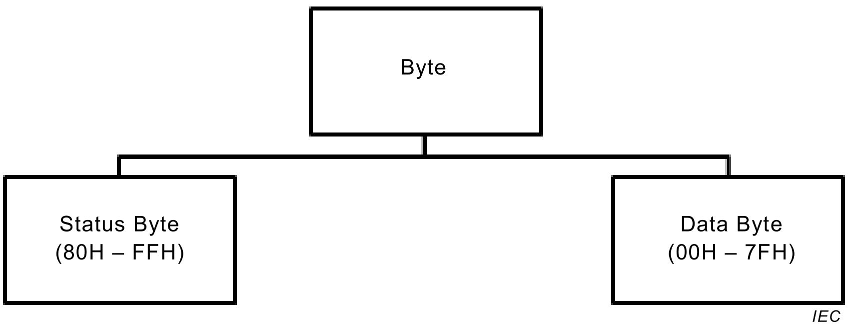


Figure 2 – Types of MIDI bytes

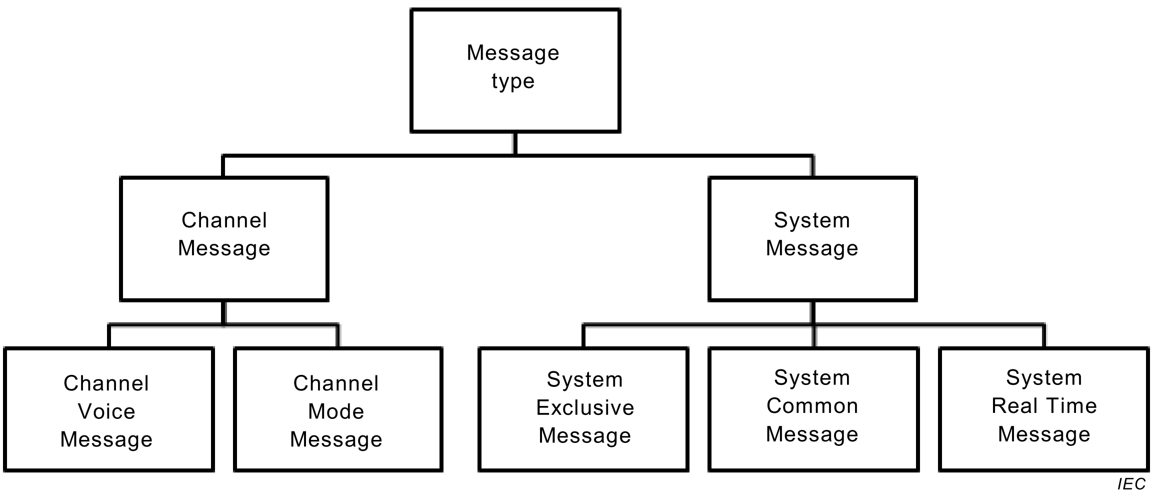


Figure 3 – Types of MIDI messages

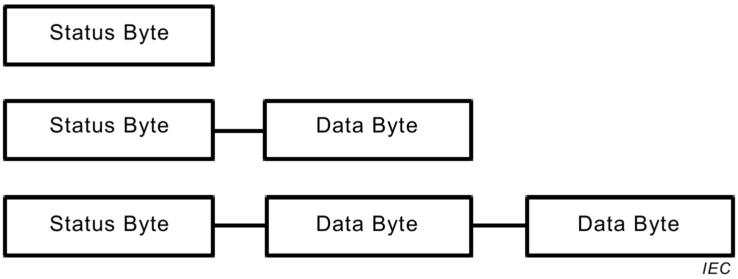


Figure 4 – Structure of a single message

Figure 5 – Structure of System Exclusive message

4.3 Message types

4.3.1 General

Messages are divided into two main categories: Channel and System.

4.3.2 Channel messages

A Channel message uses four bits in the Status byte to address the message to one of sixteen MIDI channels and four bits to define the message (see Annex A). Channel messages are thereby intended for the receivers in a system whose channel number matches the channel number encoded into the Status byte.

An instrument can receive MIDI messages on more than one channel. The channel in which it receives its main instructions, such as which program number to be on and what mode to be in, is referred to as its "Basic Channel". An instrument may be set up to receive performance data on multiple channels (including the Basic Channel). These are referred to as "Voice Channels". These multiple-channel situations will be discussed in more detail later (see 4.5).

There are two types of Channel messages: Voice and Mode.

- VOICE: To control an instrument's voices, Voice messages are sent over the Voice Channels.
- MODE: To define the instrument's response to Voice messages, Mode messages are sent over an instrument's Basic Channel.

4.3.3 System messages

System messages are not encoded with channel numbers. There are three types of System messages: Common, Real-Time, and Exclusive.

- COMMON: Common messages are intended for all receivers in a system regardless of channel.
- REAL-TIME: Real-Time messages are used for synchronization and are intended for all clock-based instruments in a system. They contain Status bytes only – no Data bytes. Real-Time messages may be sent at any time – even between bytes of a message which has a different status. In such cases the Real-Time message is either acted upon or ignored, after which the receiving process resumes under the previous status.
- EXCLUSIVE: Exclusive messages can contain any number of Data bytes, and can be terminated either by an End of Exclusive (EOX) or any other Status byte (except Real Time messages). An EOX should always be sent at the end of a System Exclusive

message. These messages include a Manufacturer's Identification (ID) code. If a receiver does not recognize the ID code, it should ignore the following data.

So that other users and third party developers can fully access their instruments, manufacturers shall publish the format of the System Exclusive data following their ID code. Only the manufacturer can define or update the format following their ID.

4.4 Data types

4.4.1 General

There are two types of bytes sent over MIDI: Status Bytes and Data bytes.

4.4.2 Status bytes

4.4.2.1 General

Status bytes are eight-bit binary numbers in which the Most Significant Bit (MSB) is set (binary 1). Status bytes serve to identify the message type, that is, the purpose of the Data bytes which follow it. Except for Real-Time messages, new Status bytes will always command a receiver to adopt a new status, even if the last message was not completed.

4.4.2.2 Running status

For Voice and Mode messages only. When a Status byte is received and processed, the receiver will remain in that status until a different Status byte is received. Therefore, if the same Status byte would be repeated, it can optionally be omitted so that only the Data bytes need to be sent. Thus, with Running Status, a complete message can consist of only Data bytes.

Running Status is especially helpful when sending long strings of Note On/Off messages, where "Note On with Velocity of 0" is used for Note Off.

Running Status will be stopped when any other Status byte intervenes. Real-Time messages should not affect Running Status.

4.4.2.3 Unimplemented status

Any status bytes, and subsequent data bytes, received for functions not implemented in a receiver should be ignored.

4.4.2.4 Undefined status

All MIDI instruments should be careful to never send any undefined status bytes. If an instrument receives any such code, it should be ignored without causing any problems to the system. Care should also be taken during power-up and power-down that no messages be sent out the MIDI Out port. Such noise, if it appears on a MIDI line, could cause a data or framing error if the number of bits in the byte is incorrect.

4.4.3 Data bytes

Following a Status byte (except for Real-Time messages) there are either one or two Data bytes which carry the content of the message. Data bytes are eight-bit binary numbers in which the Most Significant Bit (MSB) is always set to binary 0. The number and range of Data bytes which shall follow each Status byte are specified in Annex A and Annex B. For each Status byte the correct number of Data bytes shall always be sent. Inside a receiver, action on the message should wait until all Data bytes required under the current status are received. Receivers should ignore Data bytes which have not been properly preceded by a valid Status byte (with the exception of "Running Status," explained above).

4.5 Channel modes

Synthesizers and other instruments contain sound generation elements called voices. Voice assignment is the algorithmic process of routing Note On/Off data from incoming MIDI messages to the voices so that notes are correctly sounded.

NOTE When we refer to an "instrument", please note that it is possible for one physical instrument to act as several virtual instruments (i.e. a synthesizer set to a 'split' mode operates like two individual instruments). Here, "instrument" refers to a virtual instrument and not necessarily one physical instrument.

Four Mode messages are available for defining the relationship between the sixteen MIDI channels and the instrument's voice assignment. The four modes are determined by the properties Omni (On/Off), Poly, and Mono. Poly and Mono are mutually exclusive, i.e., Poly disables Mono, and vice versa. Omni, when on, enables the receiver to receive Voice messages on all voice Channels. When Omni is off, the receiver will accept Voice messages from only selected Voice Channel(s). Mono, when on, restricts the assignment of Voices to just one voice per Voice Channel (Monophonic.) When Mono is off (Poly On), a number of voices may be allocated by the Receiver's normal voice assignment (Polyphonic) algorithm.

For a receiver assigned to Basic Channel "N," (1 to 16) the four possible modes arising from the two Mode messages are shown in Table 1.

Table 1 – Modes for receiver

Mode	Omni	Poly/Mono	Description
1	On	Poly	Voice messages are received from all Voice channels and assigned to voices polyphonically.
2	On	Mono	Voice messages are received from all Voice Channels, and control only one voice, monophonically.
3	Off	Poly	Voice messages are received in Voice channel N only, and are assigned to voices polyphonically.
4	Off	Mono	Voice messages are received in Voice channels N through N+M-1, and assigned monophonically to voices 1 through M, respectively. The number of voices "M" is specified by the third byte of the Mono Mode Message.

Four modes are applied to transmitters (also assigned to Basic Channel N) (see Table 2). Transmitters with no channel selection capability should transmit on Basic Channel 1 (N=1).

Table 2 – Modes for transmitter

Mode	Omni	Poly/Mono	Description
1	On	Poly	All voice messages are transmitted in Channel N.
2	On	Mono	Voice messages for one voice are sent in Channel N.
3	Off	Poly	Voice messages for all voices are sent in Channel N.
4	Off	Mono	Voice messages for voices 1 through M are transmitted in Voice Channels N through N+M-1, respectively. (Single voice per channel).

A MIDI receiver or transmitter operates under only one Channel Mode at a time. If a mode is not implemented on the receiver, it should ignore the message (and any subsequent data bytes), or switch to an alternate mode, usually Mode 1 (Omni On/Poly).

Mode messages will be recognized by a receiver only when received in the instrument's Basic Channel – regardless of which mode the receiver is currently assigned to. Voice messages may be received in the Basic Channel and in other Voice Channels, according to the above specifications.

Since a single instrument may function as multiple "virtual" instruments, it can thus have more than one basic channel. Such an instrument behaves as though it is more than one receiver, and each receiver can be set to a different Basic Channel. Each of these receivers may also be set to a different mode, either by front panel controls or by Mode messages received over MIDI on each basic channel. Although not a true MIDI mode, instruments operating in this fashion are described as functioning in "Multi Mode."

An instrument's transmitter and receiver may be set to different modes. For example, an instrument may receive in Mono mode and transmit in Poly mode. It is also possible to transmit and receive on different channels. For example, an instrument may receive on Channel 1 and transmit on Channel 3.

4.6 Power-up default conditions

It is recommended that at power-up, the basic channel should be set to 1, and the mode set to Omni On/Poly (Mode 1). This, and any other default conditions for the particular instrument, should be maintained indefinitely (even when powered down) until instrument panel controls are operated or MIDI data is received. However, the decision to implement the above is left totally up to the designer.

5 MIDI implementation chart instructions

5.1 Introduction

The standard MIDI implementation chart is used as a quick reference of transmitter and receiver functions so that users can easily recognize what messages and functions are implemented in the instrument. This chart should be included in the users manual of all MIDI products. For example, if a user intends to connect two MIDI instruments, they might compare the "Transmitted" part of one instrument's chart, with the "Recognized" part of the other instrument's chart by overlapping them. For this reason, each chart should be the same size and have the same number of lines.

A chart template is provided in Table D.1.

5.2 General

a) The "[ ]" brackets at the top of the chart are used for the instrument's name such as, [LINEAR WAVETABLE SYNTHESIZER].

b) The item "Model" should be used for the model number, such as "LW-1".

c) The body of the implementation chart is divided into four columns. The first column is the specific function or item, the next two columns give information on whether the specified function is transmitted and/or received, and the fourth column is used for remarks. This last column is useful to explain anything unique to this implementation.

5.3 Function description

5.3.1 Basic Channel

- Default: Channel which is assigned when power is first applied to instrument.
- Changed: Channels which can be assigned from the instrument's front panel.

5.3.2 Mode

- Default: This is the channel mode used by a Transmitter and Receiver when power is first applied. This should be written as Mode x (where x is 1 through 4), as shown on the bottom of sheet.
- Messages: These are the mode messages which can be transmitted or received, such as OMNI ON/OFF, MONO ON, and POLY ON. MONO ON and POLY ON may be written in the short form "MONO", "POLY".

- Altered: This shows the channel modes which are not implemented by a receiver and the modes which are substituted. For example, if the receiver cannot accept "MONO ON" mode, but will switch to "OMNI ON" mode in order to receive the MIDI data, the following expression should be used: "MONO ON > OMNI ON" or "MONO > OMNI".

5.3.3 Note Number

- Note Number: The total range of transmitted or recognized notes.
- True Voice: Range of received note numbers falling within the range of true notes produced by the instrument.

5.3.4 Velocity

Note On/Note Off: Velocity is marked with an "o" for implemented or an "x" for not implemented. In the space following the "o" or "x", it may be mentioned how the Note On or Off data is being transmitted.

5.3.5 Aftertouch

"o" for implemented or "x" for not implemented.

5.3.6 Pitch Bend

"o" for implemented or "x" for not implemented.

5.3.7 Control Change

Space is given in this area for listing of any implemented control numbers. An "o" or "x" should be placed in the appropriate Transmitted or Recognized column, and the function of the specified control number should be listed in the Remarks column.

5.3.8 Program Change

"o" for implemented or an "x" for not implemented. If implemented, the range of numbers should be included.

True # (Number): The range of the program change numbers which correspond to the actual number of patches selected.

5.3.9 System Exclusive

"o" for implemented or "x" for not implemented. A full description of the instrument's System Exclusive implementation should be included on separate pages.

5.3.10 System Common

"o" for implemented or "x" for not implemented. The following abbreviations are used:

- Song Pos = Song Position
- Song Sel = Song Select
- Tune = Tune Request

5.3.11 System Real Time

"o" for implemented or "x" for not implemented. The following abbreviations are used:

- Clock = Timing Clock
- Commands = Start, Continue and Stop

5.3.12 Aux. messages

"o" for implemented or an "x" for not implemented. The following abbreviations are used:

- Aux = Auxiliary
- Active Sense = Active Sensing

5.3.13 Notes

The "Notes" column can be any comments such as:

Power Up messages transmitted, implementation of program changes to additional memory banks, etc.

Annex A
(normative)

Summary of MIDI messages

The following table lists many of the major MIDI messages in numerical (binary) order. Table A.1 is intended as an overview of MIDI. Additional messages and their detailed explanations are listed in The Complete MIDI 1.0 Detailed Specification and in the Confirmation of Approvals (see Bibliography).

Some Status Numbers are reserved for future extension. Those numbers should not be used in any product that is promoted as being "MIDI compatible" until defined by MMA/AMEI.

Table A.1 – MIDI Specification 1.0 message summary

Status **D7----D0**	**Data Byte(s)** **D7----D0**	**Description**
Channel Voice Messages [nnnn = 0 to 15 (MIDI Channel Number 1 to 16)]		
1000nnnn	0kkkkkkk 0vvvvvvv	Note Off event. This message is sent when a note is released (ended). (kkkkkkk) is the key (note) number. (vvvvvvv) is the velocity.
1001nnnn	0kkkkkkk 0vvvvvvv	Note On event. This message is sent when a note is depressed (start). (kkkkkkk) is the key (note) number. (vvvvvvv) is the velocity. Vvvvvvv = 0: Note Off (with velocity of 64)
1010nnnn	0kkkkkkk 0vvvvvvv	Polyphonic Key Pressure (Aftertouch). This message is most often sent by pressing down on the key after it "bottoms out". (kkkkkkk) is the key (note) number. (vvvvvvv) is the pressure value.
1011nnnn	0ccccccc 0vvvvvvv	Control Change. This message is sent when a controller value changes. Controllers include devices such as pedals and levers. Controller numbers 120 to 127 are reserved as "Channel Mode Messages" (below). (ccccccc) is the controller number (0 to 119). (vvvvvvv) is the controller value (0 to 127). (see Annex B)
1100nnnn	0ppppppp	Program Change. This message is sent when the patch number changes. (ppppppp) is the new program number.
1101nnnn	0vvvvvvv	Channel Pressure (Aftertouch). This message is most often sent by pressing down on the key after it "bottoms out". This message is different from Polyphonic Key Pressure (Aftertouch). Use this message to send the single greatest pressure value (of all the current depressed keys). (vvvvvvv) is the pressure value.
1110nnnn	0lllllll 0mmmmmmm	Pitch Wheel Change. This message is sent to indicate a change in the pitch wheel. The pitch wheel is measured by a fourteen bit value. Center (no pitch change) is 2000 H. Sensitivity is a function of the transmitter. (lllllll) are the least significant 7 bits. (mmmmmmm) are the most significant 7 bits.
Channel Mode Messages		
1011nnnn	0ccccccc 0vvvvvvv	Channel Mode Messages. This is the same Status Byte value as the Control Change, but

<table>
<tr><th>Status
D7----D0</th><th>Data Byte(s)
D7----D0</th><th>Description</th></tr>
<tr><td></td><td></td><td>implements Mode control and special message by using reserved controller numbers 120 to 127. The commands are:
– All Sound Off. When All Sound Off is received, all oscillators will turn off, and their volume envelopes are set to zero as soon as possible.
• c = 120, v = 0: All Sound Off
– Reset All Controllers. When Reset All Controllers is received, all controller values are reset to their default values. (see Bibliography, RP-015 for defaults and specific Recommended Practices for other specified applications).
• c = 121, v = x: Value shall only be zero unless otherwise allowed in a specific Recommended Practice.
– Local Control. When Local Control is Off, all instruments on a given channel will respond only to data received over MIDI. Played data, etc. will be ignored. Local Control On restores the functions of the normal controllers.
• c = 122, v = 0: Local Control Off
• c = 122, v = 127: Local Control On
– All Notes Off. When an All Notes Off is received, all oscillators will turn off.
• c = 123, v = 0: All Notes Off
• c = 124, v = 0: Omni Mode Off
• c = 125, v = 0: Omni Mode On
• c = 126, v = M: Mono Mode On (Poly Off) where M is the number of channels (Omni Off) or 0 (Omni On)
• c = 127, v = 0: Poly Mode On (Mono Off)
(Note: c = 124 to 127 also cause All Notes Off)</td></tr>
<tr><td colspan="3">System Common Messages</td></tr>
<tr><td>11110000</td><td>0iiiiiii
[0iiiiiii
0iiiiiii]
0ddddddd

0ddddddd
11110111</td><td>System Exclusive.
This message type allows manufacturers to create their own messages (such as bulk dumps, patch parameters, and other non-spec data) and provides a mechanism for creating additional MIDI Specification messages. The Manufacturer's ID code (assigned by MMA or AMEI) is either 1 byte (0iiiiiii) or 3 bytes (0iiiiiii 0iiiiiii 0iiiiiii). Two of the 1 Byte IDs are reserved for extensions called Universal Exclusive Messages, which are not manufacturer-specific. If a instrument recognizes the ID code as its own (or as a supported Universal message) it will listen to the rest of the message (0ddddddd). Otherwise, the message will be ignored. (Real-Time messages only may be interleaved with a System Exclusive.) (see Annex C)</td></tr>
<tr><td>11110001</td><td>0nnndddd</td><td>MIDI Time Code Quarter Frame.
nnn = Message Type
dddd = Values</td></tr>
<tr><td>11110010</td><td>0lllllll
0mmmmmmm</td><td>Song Position Pointer.
This is an internal 14 bit register that holds the number of MIDI beats (1 beat= six MIDI clocks) since the start of the song. (lllllll) are the least significant 7 bits. (mmmmmmm) are the most significant 7 bits.</td></tr>
<tr><td>11110011</td><td>0sssssss</td><td>Song Select.
The Song Select specifies which sequence or song is to be played.</td></tr>
<tr><td>11110100</td><td></td><td>(Reserved)</td></tr>
<tr><td>11110101</td><td></td><td>(Reserved)</td></tr>
<tr><td>11110110</td><td></td><td>Tune Request.
Upon receiving a Tune Request, all analog synthesizers should tune their oscillators.</td></tr>
</table>

Status D7----D0	Data Byte(s) D7----D0	Description
11110111		End of Exclusive. Used to terminate a System Exclusive dump (see Annex C).
System Real-Time Messages		
11111000		Timing Clock. Sent 24 times per quarter note when synchronization is required.
11111001		(Reserved)
11111010		Start. Start the current sequence playing from the beginning. (This message will be followed with Timing Clocks).
11111011		Continue. Continue at the point the sequence was Stopped.
11111100		Stop. Stop the current sequence.
11111101		(Reserved)
11111110		Active Sensing. This message is intended to be sent repeatedly to tell the receiver that a connection is alive. Use of this message is optional. When initially received, the receiver will expect to receive another Active Sensing message each 300 ms (max), and if it does not then it will assume that the connection has been terminated. At termination, the receiver will turn off all voices and return to normal (non- active sensing) operation.
11111111		Reset. Reset all receivers in the system to power-up status. This should be used sparingly, preferably under manual control. In particular, it should not be sent on power-up.

Annex B
(normative)

Control Change messages (Data bytes)

B.1 Control Change messages and Channel Mode messages

Table B.1 lists all currently defined Control Change messages and Channel Mode messages, in control number order.

Registered Parameter Numbers (RPNs) are an extension to the Control Change message for setting additional parameters. Table B.2 lists all currently defined RPNs.

Some Control Numbers and Parameter Numbers are reserved for future extension.

Table B.1 – Control Changes and Mode Changes (Status bytes 176 to 191)

Control Number (2nd Byte Value)			Control Function	3rd Byte Value	
Decimal	Binary	Hex		Value	Used As
0	00000000	00	Bank Select	0 to 127	MSB
1	00000001	01	Modulation Wheel or Lever	0 to 127	MSB
2	00000010	02	Breath Controller	0 to 127	MSB
3	00000011	03	(Reserved)	0 to 127	MSB
4	00000100	04	Foot Controller	0 to 127	MSB
5	00000101	05	Portamento Time	0 to 127	MSB
6	00000110	06	Data Entry MSB	0 to 127	MSB
7	00000111	07	Channel Volume (formerly Main Volume)	0 to 127	MSB
8	00001000	08	Balance	0 to 127	MSB
9	00001001	09	(Reserved)	0 to 127	MSB
10	00001010	0A	Pan	0 to 127	MSB
11	00001011	0B	Expression Controller	0 to 127	MSB
12	00001100	0C	Effect Control 1	0 to 127	MSB
13	00001101	0D	Effect Control 2	0 to 127	MSB
14	00001110	0E	(Reserved)	0 to 127	MSB
15	00001111	0F	(Reserved)	0 to 127	MSB
16	00010000	10	General Purpose Controller 1	0 to 127	MSB
17	00010001	11	General Purpose Controller 2	0 to 127	MSB
18	00010010	12	General Purpose Controller 3	0 to 127	MSB
19	00010011	13	General Purpose Controller 4	0 to 127	MSB
20 to 31	00010100 to 00011111	14 to 1F	(Reserved)	0 to 127	MSB
32	00100000	20	LSB for Control 0 (Bank Select)	0 to 127	LSB
33	00100001	21	LSB for Control 1 (Modulation Wheel or Lever)	0 to 127	LSB
34	00100010	22	LSB for Control 2 (Breath Controller)	0 to 127	LSB
35	00100011	23	LSB for Control 3 (Reserved)	0 to 127	LSB

Control Number (2nd Byte Value)			Control Function	3rd Byte Value	
Decimal	Binary	Hex		Value	Used As
36	00100100	24	LSB for Control 4 (Foot Controller)	0 to 127	LSB
37	00100101	25	LSB for Control 5 (Portamento Time)	0 to 127	LSB
38	00100110	26	LSB for Control 6 (Data Entry)	0 to 127	LSB
39	00100111	27	LSB for Control 7 (Channel Volume, formerly Main Volume)	0 to 127	LSB
40	00101000	28	LSB for Control 8 (Balance)	0 to 127	LSB
41	00101001	29	LSB for Control 9 (Reserved)	0 to 127	LSB
42	00101010	2A	LSB for Control 10 (Pan)	0 to 127	LSB
43	00101011	2B	LSB for Control 11 (Expression Controller)	0 to 127	LSB
44	00101100	2C	LSB for Control 12 (Effect control 1)	0 to 127	LSB
45	00101101	2D	LSB for Control 13 (Effect control 2)	0 to 127	LSB
46	00101110	2E	LSB for Control 14 (Reserved)	0 to 127	LSB
47	00101111	2F	LSB for Control 15 (Reserved)	0 to 127	LSB
48	00110000	30	LSB for Control 16 (General Purpose Controller 1)	0 to 127	LSB
49	00110001	31	LSB for Control 17 (General Purpose Controller 2)	0 to 127	LSB
50	00110010	32	LSB for Control 18 (General Purpose Controller 3)	0 to 127	LSB
51	00110011	33	LSB for Control 19 (General Purpose Controller 4)	0 to 127	LSB
52	00110100	34	LSB for Control 20 (Reserved)	0 to 127	LSB
53	00110101	35	LSB for Control 21 (Reserved)	0 to 127	LSB
54	00110110	36	LSB for Control 22 (Reserved)	0 to 127	LSB
55	00110111	37	LSB for Control 23 (Reserved)	0 to 127	LSB
56	00111000	38	LSB for Control 24 (Reserved)	0 to 127	LSB
57	00111001	39	LSB for Control 25 (Reserved)	0 to 127	LSB
58	00111010	3A	LSB for Control 26 (Reserved)	0 to 127	LSB
59	00111011	3B	LSB for Control 27 (Reserved)	0 to 127	LSB
60	00111100	3C	LSB for Control 28 (Reserved)	0 to 127	LSB
61	00111101	3D	LSB for Control 29 (Reserved)	0 to 127	LSB
62	00111110	3E	LSB for Control 30 (Reserved)	0 to 127	LSB
63	00111111	3F	LSB for Control 31 (Reserved)	0 to 127	LSB
64	01000000	40	Damper Pedal On/Off (Sustain)	≤ 63 off, ≥ 64 on	---
65	01000001	41	Portamento On/Off	≤ 63 off, ≥ 64 on	---
66	01000010	42	Sostenuto On/Off	≤ 63 off, ≥ 64 on	---
67	01000011	43	Soft Pedal On/Off	≤ 63 off, ≥ 64 on	---
68	01000100	44	Legato Footswitch	≤ 63 Normal, ≥ 64 Legato	---
69	01000101	45	Hold 2	≤ 63 off, ≥ 64 on	---
70	01000110	46	Sound Controller 1 (default: Sound Variation)	0 to 127	---
71	01000111	47	Sound Controller 2 (default: Timbre/Harmonic Intensity)	0 to 127	---
72	01001000	48	Sound Controller 3 (default: Release Time)	0 to 127	---

Control Number (2nd Byte Value)			Control Function	3rd Byte Value	
Decimal	Binary	Hex		Value	Used As
73	01001001	49	Sound Controller 4 (default: Attack Time)	0 to 127	---
74	01001010	4A	Sound Controller 5 (default: Brightness)	0 to 127	---
75	01001011	4B	Sound Controller 6 (default: Decay Time – see Bibliography, RP-021)	0 to 127	---
76	01001100	4C	Sound Controller 7 (default: Vibrato Rate – see Bibliography, RP-021)	0 to 127	---
77	01001101	4D	Sound Controller 8 (default: Vibrato Depth – see Bibliography, RP-021)	0 to 127	---
78	01001110	4E	Sound Controller 9 (default: Vibrato Delay – see Bibliography, RP-021)	0 to 127	---
79	01001111	4F	Sound Controller 10 (default undefined – see Bibliography, RP-021)	0 to 127	---
80	01010000	50	General Purpose Controller 5	0 to 127	---
81	01010001	51	General Purpose Controller 6	0 to 127	---
82	01010010	52	General Purpose Controller 7	0 to 127	---
83	01010011	53	General Purpose Controller 8	0 to 127	---
84	01010100	54	Portamento Control	0 to 127	---
85	01010101	55	(Reserved)	---	---
86	01010110	56	(Reserved)	---	---
87	01010111	57	(Reserved)	---	---
88	01011000	58	High Resolution Velocity Prefix	0 to 127	---
89	01011001	59	(Reserved)	---	---
90	01011010	5A	(Reserved)	---	---
91	01011011	5B	Effects 1 Depth (default: Reverb Send Level – see Bibliography, RP-023) (formerly External Effects Depth)	0 to 127	---
92	01011100	5C	Effects 2 Depth (formerly Tremolo Depth)	0 to 127	---
93	01011101	5D	Effects 3 Depth (default: Chorus Send Level – see Bibliography, RP-023) (formerly Chorus Depth)	0 to 127	---
94	01011110	5E	Effects 4 Depth (formerly Celeste [Detune] Depth)	0 to 127	---
95	01011111	5F	Effects 5 Depth (formerly Phaser Depth)	0 to 127	---
96	01100000	60	Data Increment (Data Entry +1) (see Bibliography, RP-018)	N/A	---
97	01100001	61	Data Decrement (Data Entry -1) (see Bibliography, RP-018)	N/A	---
98	01100010	62	Non-Registered Parameter Number (NRPN) – LSB	0 to 127	LSB
99	01100011	63	Non-Registered Parameter Number (NRPN) – MSB	0 to 127	MSB
100	01100100	64	Registered Parameter Number (RPN) – LSB (see Annex B.2)	0 to 127	LSB
101	01100101	65	Registered Parameter Number (RPN) – MSB (see Annex B.2)	0 to 127	MSB

Control Number (2nd Byte Value)			Control Function	3rd Byte Value	
Decimal	Binary	Hex		Value	Used As
102 to 119	01100110 to 01110111	66 to 77	(Reserved)	---	---
120	01111000	78	[Channel Mode Message] All Sound Off	0	---
121	01111001	79	[Channel Mode Message] Reset All Controllers (see Bibliography, RP-015)	0	---
122	01111010	7A	[Channel Mode Message] Local Control On/Off	0 off, 127 on	---
123	01111011	7B	[Channel Mode Message] All Notes Off	0	---
124	01111100	7C	[Channel Mode Message] Omni Mode Off (+ all notes off)	0	---
125	01111101	7D	[Channel Mode Message] Omni Mode On (+ all notes off)	0	---
126	01111110	7E	[Channel Mode Message] Mono Mode On (+ poly off, + all notes off)	Note: This equals the number of channels, or zero if the number of channels equals the number of voices in the receiver.	---
127	01111111	7F	[Channel Mode Message] Poly Mode On (+ mono off, +all notes off)	0	---
NOTE Controller numbers 120 to 127 are reserved for Channel Mode Messages, which rather than controlling sound parameters, affect the channel's operating mode (see Annex A).					

B.2 Registered Parameter numbers

To set or change the value of a Registered Parameter:

a) Send two Control Change messages using Control Numbers 101 (65H) and 100 (64H) to select the desired Registered Parameter Number, as per Table B.2.

b) To set the selected Registered Parameter to a specific value, send a Control Change messages to the Data Entry MSB controller (Control Number 6). If the selected Registered Parameter requires the LSB to be set, send another Control Change message to the Data Entry LSB controller (Control Number 38).

c) To make a relative adjustment to the selected Registered Parameter's current value, use the Data Increment or Data Decrement controllers (Control Numbers 96 and 97).

Table B.2 – Registered Parameter numbers

Parameter Number		Parameter Function	Data Entry Value
MSB: Control 101 (65H) Value	**LSB: Control 100 (64H) Value**		
00H	00H	Pitch Bend Sensitivity	MSB = +/- semitones [a] LSB =+/- cents [b]
	01H	Channel Fine Tuning (formerly Fine Tuning – see Bibliography, RP-022)	Resolution **Error!** cents [b] – 00H 00H = –100 cents [b] – 40H 00H = A440 – 7FH 7FH = +100 cents [b]
	02H	Channel Coarse Tuning (formerly Coarse Tuning – see Bibliography, RP-022)	Only MSB used Resolution 100 cents [b] – 00H = –6400 cents [b] – 40H = A440 – 7FH = +6300 cents [b]
	03H	Tuning Program Change	Tuning Program Number
	04H	Tuning Bank Select	Tuning Bank Number
	05H	Modulation Depth Range (see Bibliography, RP-024)	For General MIDI Level 2, see Bibliography, RP-024 For other systems, defined by manufacturer
	06H to 7FH	(Reserved)	
01H to 3CH	(Reserved)		
3DH	Three Dimensional Sound Controllers		
	00H	Azimuth Angle	see Bibliography, RP-049
	01H	Elevation Angle	see Bibliography, RP-049
	02H	Gain	see Bibliography, RP-049
	03H	Distance Ratio	see Bibliography, RP-049
	04H	Maximum Distance	see Bibliography, RP-049
	05H	Gain at Maximum Distance	see Bibliography, RP-049
	06H	Reference Distance Ratio	see Bibliography, RP-049
	07H	Pan Spread Angle	see Bibliography, RP-049
	08H	Roll Angle	see Bibliography, RP-049
	09H to 7FH	(Reserved)	
3EH to 7EH	(Reserved)		
7FH	Null Function Number for RPN/NRPN		Setting RPN to 7FH will disable the data entry, data increment, and data decrement controllers until a new RPN or NRPN is selected.
NOTE			
[a] semitone: 1 semitone = **Error!** oct [b] cent: 1 cent = **Error!** oct			

Annex C
(normative)

System Exclusive messages

C.1 System Exclusive messages

Table C.1 shows the format of System Exclusive messages.

Table C.1 – System Exclusive messages

Status		Data Bytes	Description	
Hex	Binary			
F0H	11110000		SOX: Start of System Exclusive Status Byte	
		0iiiiiii	System Exclusive ID number [a]	
			(00 to 7CH)	Manufacturer Identification
			(7DH)	Non Commercial System Exclusive ID
			(7EH)	Non-Real Time System Exclusive
			(7FH)	Real Time System Exclusive
		0ddddddd . . 0ddddddd	Any number of data bytes may be sent here, for any purpose, as long as they all have a zero in the most significant bit [b]	
F7H	11110111		EOX: End of System Exclusive	

[a] – Manufacturer identification (0 to 124). If the first byte of this ID is 0, the following two bytes are used as extensions to the Manufacturer ID. A Manufacturer ID may be obtained from the MIDI Manufacturers Association and the Association of Musical Electronics Industry for Japanese manufacturers.

– ID 7DH (125) is reserved for non-commercial use (e.g. schools, research, etc.) and is not to be used on any product released to the public.

– ID 7EH (126) and 7FH (127) are used for Universal System Exclusive extensions to the MIDI specification. See Table C.2 for a listing of currently defined Non-Real Time and Real Time messages.

[b] All bytes between the System Exclusive Status byte and EOX shall have zeroes in the Most Significant Bit – which therefore makes them Data Bytes – with the exception of System Real Time Status Bytes (F8H to FFH) (see Annex A). Any other Status Byte that appears between the SOX (F0H) and EOX (F7H) will terminate the System Exclusive message.

C.2 Universal System Exclusive messages

Table C.2 lists all currently defined Universal System Exclusive Messages.

Universal System Exclusive Messages are defined as Real Time or Non-Real Time, and are used for extensions to MIDI that are not intended to be manufacturer exclusive (despite the name).

Many of these messages are defined in The Complete MIDI 1.0 Detailed Specification and Confirmation of Approvals (see Bibliography). Others are defined in Recommended Practice documentation (see Bibliography).

Some Sub-ID Numbers are reserved for future extension.

Table C.2 – Universal System Exclusive messages

Non-Real Time (7EH)			
Sub-ID 1	**Sub-ID 2**	**Description**	
00		(Reserved)	
01		Sample Dump Header	
02		Sample Data Packet	
03		Sample Dump Request	
04	nn	MIDI Time Code	
	00		Special
	01		Punch In Points
	02		Punch Out Points
	03		Delete Punch In Point
	04		Delete Punch Out Point
	05		Event Start Point
	06		Event Stop Point
	07		Event Start Points with additional info.
	08		Event Stop Points with additional info.
	09		Delete Event Start Point
	0A		Delete Event Stop Point
	0B		Cue Points
	0C		Cue Points with additional info.
	0D		Delete Cue Point
	0E		Event Name in additional info.
	0F to 7F		(Reserved)
05	nn	Sample Dump Extensions	
	00		(Reserved)
	01		Loop Points Transmission
	02		Loop Points Request
	03		Sample Name Transmission
	04		Sample Name Request
	05		Extended Dump Header
	06		Extended Loop Points Transmission
	07		Extended Loop Points Request
	08 to 7F		(Reserved)
06	nn	General Information	
	00		(Reserved)
	01		Identity Request
	02		Identity Reply
	03 to 7F		(Reserved)
07	nn	File Dump	
	00		(Reserved)
	01		Header
	02		Data Packet
	03		Request
	04 to 7F		(Reserved)

Non-Real Time (7EH)			
Sub-ID 1	**Sub-ID 2**	**Description**	
08	nn	MIDI Tuning Standard (Non-Real Time)	
	00		Bulk Dump Request
	01		Bulk Dump Reply
	02		(Reserved)
	03		Tuning Dump Request
	04		Key-Based Tuning Dump
	05		Scale/Octave Tuning Dump, 1 byte format
	06		Scale/Octave Tuning Dump, 2 byte format
	07		Single Note Tuning Change with Bank Select
	08		Scale/Octave Tuning, 1 byte format
	09		Scale/Octave Tuning, 2 byte format
	0A to 7F		(Reserved)
09	nn	General MIDI	
	00		(Reserved)
	01		General MIDI 1 System On
	02		General MIDI System Off
	03		General MIDI 2 System On
	04 to 7F		(Reserved)
0A	nn	Downloadable Sounds	
	00		(Reserved)
	01		Turn DLS On
	02		Turn DLS Off
	03		Turn DLS Voice Allocation Off
	04		Turn DLS Voice Allocation On
	05 to 7F		(Reserved)
0B	nn	File Reference Message	
	00		(Reserved)
	01		Open File
	02		Select or Reselect Contents
	03		Open File and Select Contents
	04		Close File
	05 to 7F		(Reserved)
0C	nn	MIDI Visual Control	
	00 to 7F		MVC Commands (see Bibliography, RP-050)
0D to 7A		(Reserved)	
7B	--	End of File	
7C	--	Wait	
7D	--	Cancel	
7E	--	NAK	
7F	--	ACK	

Real Time (7FH)			
Sub-ID #1	**Sub-ID #2**	**Description**	
00		(Reserved)	
01	nn	MIDI Time Code	
	00		(Reserved)
	01		Full Message
	02		User Bits
	03 to 7F		(Reserved)
02	nn	MIDI Show Control	
	00		MSC Extensions
	01 to 7F		MSC Commands (see The Complete MIDI 1.0 Detailed Specification)
03	nn	Notation Information	
	00		(Reserved)
	01		Bar Number
	02		Time Signature (Immediate)
	03 to 41		(Reserved)
	42		Time Signature (Delayed)
	43 to 7F		(Reserved)
04	nn	Device Control	
	00		(Reserved)
	01		Master Volume
	02		Master Balance
	03		Master Fine Tuning
	04		Master Course Tuning
	05		Global Parameter Control
	06 to 7F		(Reserved)
05	nn	Real Time MTC Cueing	
	00		Special
	01		Punch In Points
	02		Punch Out Points
	03		(Reserved)
	04		(Reserved)
	05		Event Start points
	06		Event Stop points
	07		Event Start points with additional info.
	08		Event Stop points with additional info.
	09		(Reserved)
	0A		(Reserved)
	0B		Cue points
	0C		Cue points with additional info.
	0D		(Reserved)
	0E		Event Name in additional info.
	0F to 7F		(Reserved)
06	nn	MIDI Machine Control Commands	

Real Time (7FH)			
Sub-ID #1	**Sub-ID #2**	**Description**	
	00 to 7F		MMC Commands (see The Complete MIDI 1.0 Detailed Specification)
07	nn	MIDI Machine Control Responses	
	00 to 7F		MMC Responses (see The Complete MIDI 1.0 Detailed Specification)
08	nn	MIDI Tuning Standard (Real Time)	
	00 to 01		(Reserved)
	02		Single Note Tuning Change
	03 to 06		(Reserved)
	07		Single Note Tuning Change with Bank Select
	08		Scale/Octave Tuning, 1 byte format
	09		Scale/Octave Tuning, 2 byte format
	0A to 7F		(Reserved)
09	nn	Controller Destination Setting (see Bibliography, RP-024)	
	00		(Reserved)
	01		Channel Pressure (Aftertouch)
	02		Polyphonic Key Pressure (Aftertouch)
	03		Controller (Control Change)
	04 to 7F		(Reserved)
0A	00	(Reserved)	
	01	Key-Based Instrument Control	
	02 to 7F	(Reserved)	
0B	00	(Reserved)	
	01	Scalable Polyphony MIDI MIP Message	
	02 to 7F	(Reserved)	
0C	00	Mobile Phone Control Message	
	01 to 7F	(Reserved)	
0D to 7F		(Reserved)	

NOTE 1 The standardized format for both Real Time and Non-Real Time messages is as follows:

F0H <ID number> <device ID> <sub-ID#1> <sub-ID#2>…… F7H

NOTE 2 Since System Exclusive messages are not assigned to a MIDI Channel, the device ID is intended to indicate which instrument in the system is the target for this message. The device ID 7F is used to indicate that all instruments are the target. Additional details and descriptions are available in The Complete MIDI 1.0 Detailed Specification (see Bibliography).

Annex D
(normative)

MIDI Implementation Chart template

Table D.1 – MIDI Implementation Chart template

[Model] MIDI Implementation Chart — Date: Version:

Function…		Transmitted	Recognized	Remarks
Basic Channel	Default Changed			
Mode	Default Messages Altered	 **************		
Note Number	True voice	**************		
Velocity	Note On Note Off			
After Touch	Key's Channel			
Pitch Bend				
Control Change				
Program Change	True Number	**************		
System Exclusive				
System Common	Song Position Song Select Tune Request			
System Real Time	Clock Commands			
Aux Messages	Local On/Off All Notes Off Active Sensing System Reset			
Notes				

Mode 1: Omni On, Poly Mode 2: Omni On, Mono O: Yes
Mode 3: Omni Off, Poly Mode 4: Omni Off, Mono X: No

Bibliography

Details about implementing MIDI messages can dramatically impact compatibility with other products. It is strongly recommended that implementers consult the following referenced documents for additional information.

The Complete MIDI 1.0 Detailed Specification, MIDI Manufacturers Association. Available from <http://www.midi.org>. This document captures the complete definition of MIDI protocol through 1996.

Confirmation of Approvals, MIDI Manufacturers Association. Available individually from <http://www.midi.org>. These documents capture message definitions made after 1996. Currently defined Confirmations of Approvals are as follows:

- MMA-0017, *DLS-1 Universal System Exclusive messages*
- CA-018, *File Reference System Exclusive messages*
- CA-019, *Sample Dump Size, Rate and Name extensions*
- CA-020, *MIDI Tuning Bank/Dump extensions*
- CA-021, *Scale/Octave Tuning*
- CA-021/amd1, *Corrections to Scale Octave Tuning*
- CA-022, *Controller Destination System Exclusive message*
- CA-023, *Key-Based Instrument Control SysEx messages*
- CA-024, *Global Parameter Control System Exclusive message*
- CA-025, *Master Fine/Coarse Tune System Exclusive message*
- CA-026, *Modulation Depth Range RPN*
- CA-027, *General MIDI 2 System Exclusive message*
- CA-028, *Extension 00-01 to File Reference SysEx Message*
- CA-029, *Maximum Instantaneous Polyphony Message*
- CA-030, *Mobile Phone Control SysEx Message*
- CA-031, *High Resolution Velocity Prefix (CC#88)*
- CA-032, *MIDI Visual Control Universal Non Realtime SysEx Message*
- CA-033, *Updated MIDI Electrical Specification*

Recommended Practices, MIDI Manufacturers Association. Available individually from <http://www.midi.org>. These documents have been established for use of MIDI, such as features for devices and other implementation details which provide for interoperability. The following are the Recommended Practices approved to date:

- RP-001, *Standard MIDI Files*
- RP-002, *MIDI Show Control*
- RP-003, *General MIDI 1*
- RP-004, *Redefinition of MTC (MIDI Time Code) User Bits Message*
- RP-005, *Notation Information: Bar Marker*
- RP-006, *Notation Information: Time Signature*
- RP-007, *Master Volume and Balance*
- RP-008, *Realtime MTC (MIDI Time Code) Cueing*
- RP-009, *File Dump (Data Transfer SysEx)*
- RP-010, *Sound Controllers – Instrument Defaults*

- RP-011, *Sound Controllers – Effects Defaults – suspended*
- RP-012, *MIDI Tuning Standard*
- RP-013, *MMC (MIDI Machine Control)*
- RP-014, *MSC (MIDI Show Control) 1.1 (2-Phase Commit)*
- RP-015, *Response to Reset All Controllers*
- RP-016, *Downloadable Sounds (Level 1) Specification*
- RP-017, *SMF (Standard MIDI File) Lyric Event Definition*
- RP-018, *Response to Data Increment/Decrement Controller*
- RP-019, *SMF (Standard MIDI File) Device/Program Name Meta Events*
- RP-020, *Scale/Octave Tuning Defaults*
- RP-021, *Sound Controllers – Revised Defaults*
- RP-022, *Channel Fine/Coarse Tuning (Redefinition of RPN 01-02)*
- RP-023, *CC (Control Change) 91/93 Function Renaming*
- RP-024, *General MIDI 2 Specification v1.0*
- RP-025, *Downloadable Sounds (Level 2) Specification*
- RP-025/*amd1, Downloadable Sounds (Level 2.1) Specification*
- RP-025/*amd2, Downloadable Sounds (Level 2.2) Specification*
- RP-026, *SMF (Standard MIDI File) Language and Display extensions*
- RP-027, *MIDI Media Adaptation Layer for IEEE-1394 Specification*
- RP-028, *Version 2.0 MIDI Implementation Chart*
- RP-029, *Technical Note: DLS (Downloadable Sounds) & SMF (Standard MIDI File) in* RMID Fi*les*
- RP-030, *XMF Meta File Format Specification*
- RP-031, *Type 0 and 1 XMF Files*
- RP-032, *SMF (Standard MIDI File) Meta Event for XMF Patch Type Prefix*
- RP-033, *General MIDI Lite Specification (GML) and Guidelines for Mobile Applications*
- RP-034, *Scalable Polyphony MIDI*
- RP-035, *SP-MIDI (Scalable Polyphony MIDI) 5-to-24 Note Profile for 3GPP*
- RP-036, *Default Pan Curve*
- RP-037, *GM2 (General MIDI 2) MIDI Tuning Amendment*
- RP-038, *MIDI Note, Patch and Control Names in XML*
- RP-039, *XMF Meta File Format Updates 1.01*
- RP-040, *XMF Compression Definition for ZLIB*
- RP-041, *Mobile DLS (Downloadable Sounds)*
- RP-042, *Mobile XMF Content Format Specification*
- RP-043, *XMF Meta File Format 2.0*
- RP-044, *GM2 (General MIDI 2) response to Mod Depth Range RPN*
- RP-045, *Audio Clips For Mobile XMF*
- RP-046, *Specification for Mobile Phone Control*
- RP-047, *ID3 Metadata for XMF Files*
- RP-048, *Mobile Musical Instrument Specification*
- RP-049, *Three Dimensional Sound Controllers*

- RP-050, MIDI Visual Control Specification
- RP-051, *International Standard MIDI Code*
- RP-052, *Specification for MIDI over Bluetooth Low Energy (BLE-MIDI)*

___

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch